SmartCare HMS — Integrated Enterprise Hospital Management

System Architecture Specification & Technical Documentation — Category: Enterprise Full Stack Healthcare

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1. Executive Summary & Clinical Domain Overview

SmartCare HMS is a clinical-grade Hospital Management System engineered to automate end-to-end healthcare operations across outpatient (OPD) and inpatient (IPD) departments. Supporting 16 distinct role workflows, the system bridges clinical charting, electronic health records (EHR), diagnostic laboratory (LIS), radiology (RIS), pharmacy inventory, surgical operating theaters (OT), and billing.

The platform incorporates clinical decision support (CDS) heuristics, a voice-enabled assistant, and an integrated medical knowledge RAG chatbot to reduce physician burnout and administrative latency.

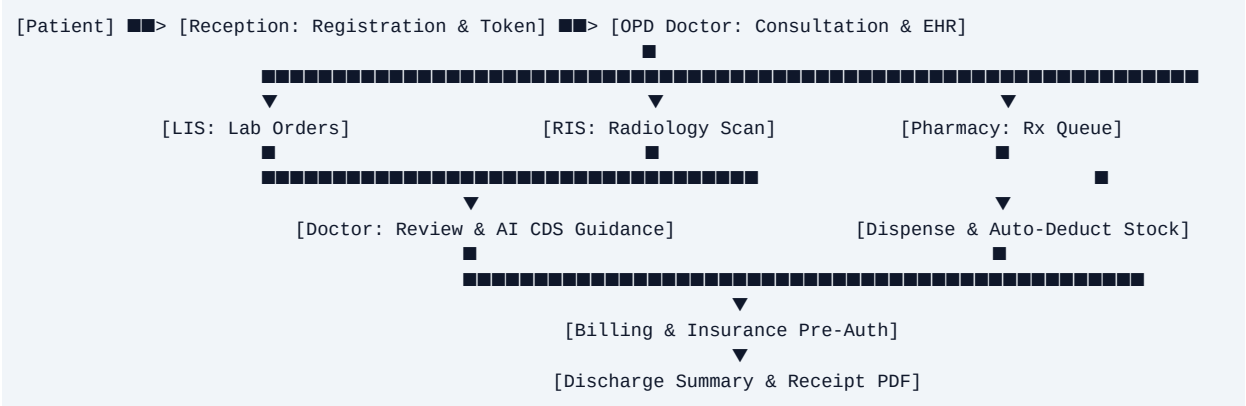
2. Supported User Roles & Modular Architecture

Role Category	Role Identities	Primary Operational Responsibilities
Clinical Care	Doctor, Nurse, OT Staff	OPD consultations, EHR charting, MAR vitals logging, surgical checklists
Diagnostics	Lab Tech, Radiologist	LIS sample collection, test verification, DICOM imaging review, PDF reports
Operations	Reception, Pharmacy, Inventory	Token queue, patient registration, IPD bed allocations, drug dispensing
Administration	Admin, HR, Billing, Insurance	User auditing, multi-department invoicing, cashless TPA claims, fleet tracking
Emergency	Ambulance Driver, Operator	Real-time dispatch, GPS telemetry, triage priority notifications

3. Technology Stack & Specifications

Subsystem	Technology	Function in Platform
Frontend Client	React 18, TypeScript, Tailwind	Type-safe reactive dashboard, role navigation, responsive data grids
Backend Framework	Python 3, Django REST Framework	Modular Django apps: clinical, diagnostics, billing, auth, inventory
Database Layer	PostgreSQL / SQLite	Relational health records, foreign keys, timestamps, audit histories
Authentication	SimpleJWT + Google OAuth	Multi-role JWT bearer tokens, demo quick-fill presets, password hashing
Document Engine	ReportLab / Canvas PDF	Diagnostic lab results, invoice receipts, clinical discharge summaries

4. Clinical Workflow & Data Flow Diagram (DFD Level 1)



5. Database Relational Architecture

• **accounts_user:** Extended user model with role-based choices, department assignments, contact data, and active credentials.

• **patients_patient:** Medical record number (MRN), date of birth, blood group, allergies, chronic conditions, and emergency contact.

• **clinical_appointment:** Doctor reference, patient reference, appointment slot, token number, triage priority, and visit status.

• **diagnostics_laborder:** Test type, sample collection timestamp, technician notes, verified values, reference ranges, and report PDF path.

• **billing_invoice:** Aggregate charges across consultations, bed days, pharmacy items, laboratory tests, and insurance coverage deduction.

6. Security, Compliance & Future Roadmap

• **Security:** Strict RBAC permission classes (`IsDoctor`, `IsReceptionist`, `IsLabTech`); PBKDF2 credential hashing; audit trails on EHR updates.

• **Testing:** Seed script verifies 16 pre-configured demo user accounts (`Password@123`) across all medical departments.

• **Future Scope:** FHIR standard interoperability, HL7 message integration, and real-time WebRTC teleconsultation streaming.

Architecture Note: Data isolation is enforced at the API layer: patients cannot view peer records; medical staff access is scoped to assigned wards and appointments.